

Pavel Marius Florian Adrian

Grupa 22

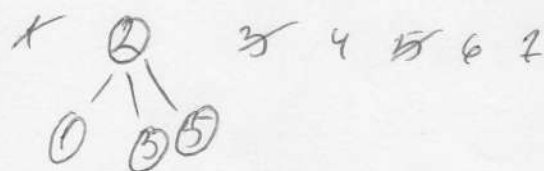
Laboratorul 8

1) ~~(1,2)2~~ $\rightarrow$ (1,2)2
 (1,3)4 $\rightarrow$ (2,5)3
 (1,4)7 $\rightarrow$ (1,3)4
 (1,6)5 $\rightarrow$ (1,6)5
 (2,4)6 $\rightarrow$ (2,4)6
 (2,5)3 $\rightarrow$ ~~(2,6)6~~
 (2,7)8 $\rightarrow$ (4,7)6
 (3,6)6 (6,7)6
 (4,6)7 (1,4)7
 (4,7)6 (4,7)7
 (5,7)7 (5,7)7
 (6,7)6 (2,7)8

1	2	3	4	5	6	7
1	1	1	-1	1	-1	1
2	2	2		2		
		-2	-4			

$$\text{Cost} = 0 + 2 + 3 + 4 = 9$$

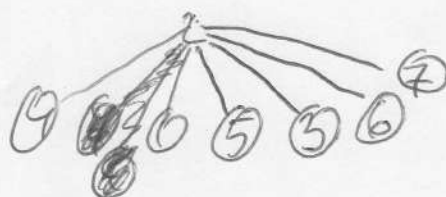
$$L = \{(1,2), (2,5), (1,3)\}$$



1	2	3	4	5	6	7
2	1	2	1	2	1	1
	1	2		2	2	
	1					
	1					
	1					

$$\text{Cost} = 5 + 5 + 6 + 6 = 26$$

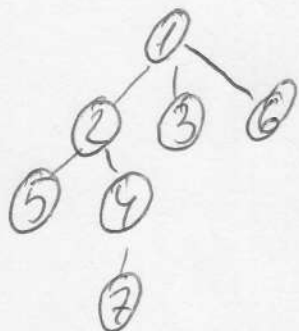
$$L = \{(1,2), (2,5), (1,3), (1,6), (2,4), (4,7)\}$$



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1) Arborele partiiale de cost minim



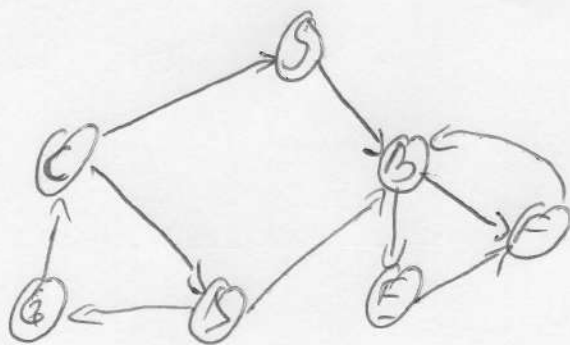
2) a) $V = \{A, B, C, D, E, F, G, S\}$

$E = \{(A, C); (A, B); (B, F); (B, F); (C, S); (C, D); (D, B); (D, C); (E, A); (E, F); (F, B); (G, C); (S, A); (S, B)\}$

$$b) \begin{cases} d^-(S) = 1 \\ d^+(S) = 2 \end{cases} \quad \begin{cases} d^-(C) = 2 \\ d^+(C) = 2 \end{cases}$$

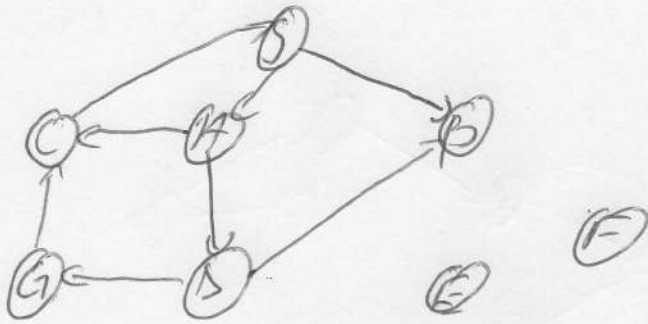
$$\begin{cases} d^-(B) = 3 \\ d^+(B) = 2 \end{cases} \quad \begin{cases} d^-(F) = 2 \\ d^+(F) = 1 \end{cases}$$

c) Subgraf eliminat modul A



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2) c) Graf partial eliminând muchiile (B,E), (E,A), (B,F) (F,B)

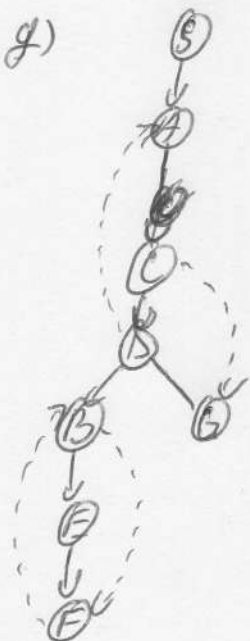
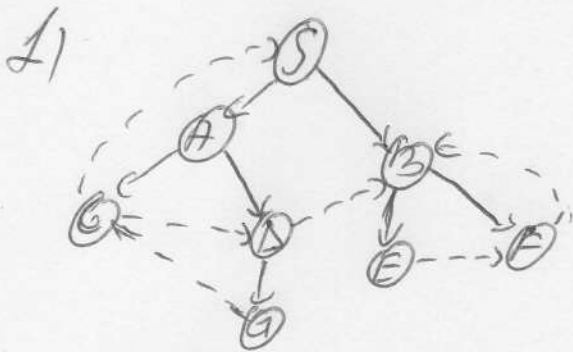


d) Drum: C, D, G, C, S, B

Drum elementor: A, C, D, B, E, F

e) Circuit: S, A, C, D, G, E, S

Circuit elementor: S, A, D, G, C, S



2) b)

	A	B	C	D	E	F	G	S
A	0	0	1	1	0	0	0	0
B	0	0	0	0	1	1	0	0
C	0	0	0	1	0	0	0	1
D	0	1	0	0	0	0	1	0
E	0	0	0	0	0	1	0	0
F	0	1	0	0	0	0	0	0
G	0	0	1	0	0	0	0	0
S	1	1	0	0	0	0	0	0

i) $A \propto C, D$ $E \propto F$
 $B \propto E, F$ $F \propto B$
 $C \propto D, S$ $G \propto C$
 $D \propto B, G$ $S \propto A, B$

3) Toate modulele au un părinte și tot. copii (modulele) sunt vecini

BFS:



BFS: Fiecare nod are un singur copil

